TASK – 1

RTL Design & Functional Simulation (Verilog)

1.Aim: To design and simulate a 4-bit binary counter using Verilog HDL and verify its functionality using a testbench.

2.Objective

To understand RTL (Register Transfer Level) design

To write Verilog code for a sequential circuit

To perform functional simulation and analyze output waveforms

3. Tools Used

- EDA Playground (Online Simulator)
- Icarus Verilog (Simulator)
- EPWave (Waveform Viewer)

4. Theory

- RTL (Register Transfer Level) design describes the digital system in terms of data flow between registers and logical operations. Verilog HDL is used to model such systems.
- A 4-bit counter is a sequential circuit that increments its output value on every clock pulse. It also includes a reset signal to initialize the output to zero.

5. Design Description

The designed circuit is a 4-bit up counter:

- It increments by 1 on every positive edge of the clock
- When reset is high, output becomes 0

6. Verilog Code (Design Module)

testbench.sv

```
// Code your testbench here
// or browse Examples
module tb;
reg clk, reset;
wire [3:0] out;

counter uut(clk, reset, out);

// Clock generation
always #5 clk = ~clk;

initial begin
    $dumpfile("dump.vcd");
    $dumpvars(0, tb);

    clk = 0;
    reset = 1;

    #10 reset = 0;
    #100 $finish;
end
endmodule
```

design.sv

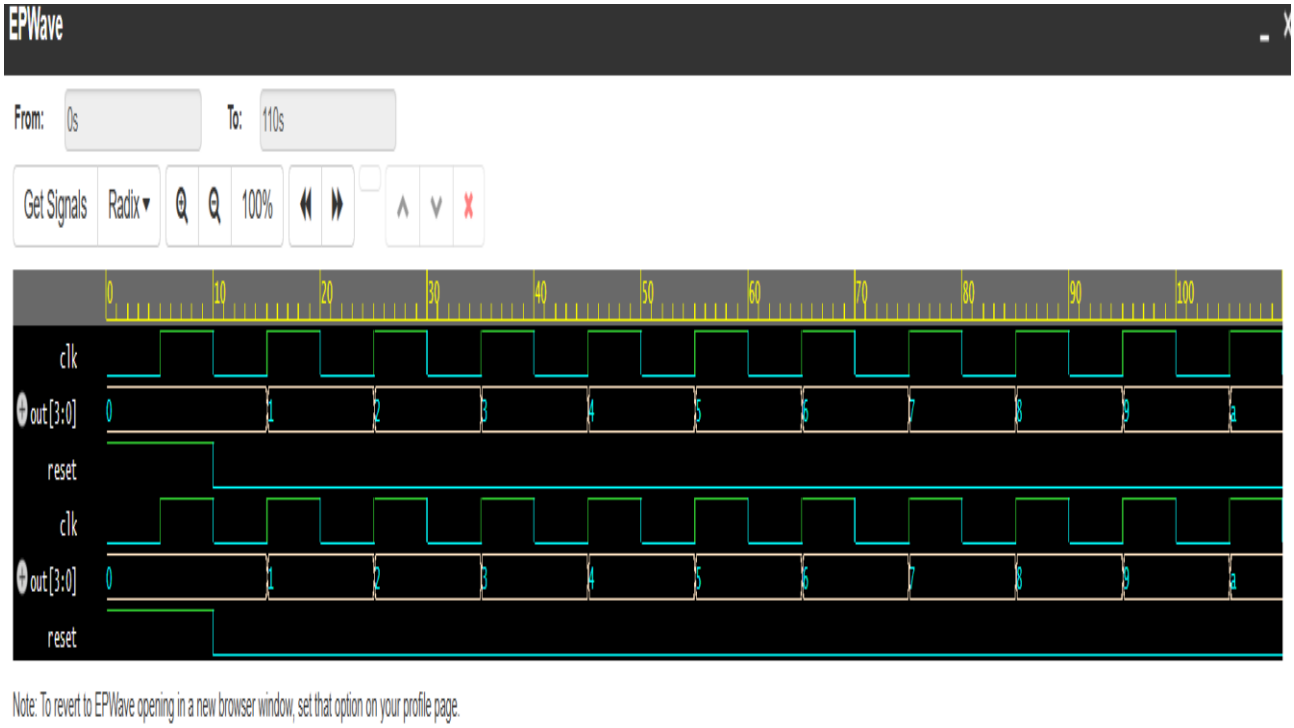
```
// Code your design here
module counter(clk, reset, out);
input clk, reset;
output reg [3:0] out;

always @(posedge clk or posedge reset)
begin
    if(reset)
        out <= 0;
    else
        out <= out + 1;
end
endmodule
```

8. Simulation Procedure

1. Open EDA Playground
2. Paste design code and testbench code
3. Select language as Verilog
4. Choose Icarus Verilog as simulator
5. Enable “Open EPWave after run”
6. Click Run
7. Observe waveform output

9. Output Waveform



10. Result

The output waveform shows that:

- The counter starts from 0 when reset is active
- It increments sequentially (0, 1, 2, 3, …)
- Output changes occur at the positive edge of the clock

Thus, the 4-bit counter is successfully designed and verified.

11. Conclusion

The RTL design of a 4-bit counter was successfully implemented using Verilog HDL. Functional simulation was performed, and correct output waveforms were obtained, validating the design.